Juan Arguello

Atlanta, GA | 678.761.5723 | jarguello8@gatech.edu | www.linkedin.com/in/juan-a-arguello | U.S Citizen

Education

Georgia Institute of Technology | Atlanta, GA *May 2026 – Present*
Bachelor of Science in Electrical Engineering, GPA 4.00 Expected Graduation, May 2028

University of Georgia | Athens, GA *August 2024 – May 2026*
Transfer with 78 Credit Hours, GPA 3.65

University of Liverpool | Liverpool, UK *June – August 2025*
Study Abroad | Supply Chain Management, Sustainable Development, Sustainability and Business Ethics

Skills

Programming: Python, Java, C, LaTeX, MATLAB, Verilog
Hardware: Raspberry Pi, STM32, Oscilloscope, Logic Analyzer
Software: GitHub, VS Code, SolidWorks CAD, STM32Cube, Microsoft Office, CYMCAP, PLS-CADD
Professional Organizations: IEEE
Languages: English (fluent), Spanish (fluent)

Experience

CME Wire and Cable | Suwanee, GA *May – August 2026*
Engineering Intern
Global manufacturer of high-voltage wire and cable solutions, specializing in power distribution and industrial power systems.

- Engineered a Python data extraction pipeline utilizing Streamlit, Pandas, NumPy, Matplotlib to analyze performance of high voltage time tests, extracting insights that drove a $250,000 investment in quality assurance initiatives.
- Reviewed technical bids and specification sheets, ensuring incoming orders and cable designs met manufacturing compliance and standards.
- Conducted ampacity and sag calculations to validate thermal ratings and mechanical integrity under various loads.

Optimus Distribution LLC | Atlanta, GA *May – June 2025*
Engineering Intern
International distributor of power grid infrastructure focused on utility supply chain logistics.

- Analyzed and reviewed technical specifications for lighting and power lines ensuring compliance of produced concrete poles with Florida Power & Light.
- Conducted lead time estimations for the trans-Atlantic transport of spun concrete poles, helping inform project planning and scheduling.
- Assisted the international supply chain for power grid infrastructure by tracking purchase orders and coordinating with logistics partners.

Projects

Embedded Systems Design Project *January – May 2026*
Hardware Measurement & Automation
Team-based academic project focused on engineering an automated hardware test-bench facilitating data acquisition and signal processing.

- Developed a Raspberry Pi-based integrated test-bench designed for automated hardware characterization and real-time signal monitoring.
- Designed and integrated high-precision instrumentation modules, including a 5-bit SAR ADC, digital voltmeter, and ohmmeter, to support accurate data acquisition.
- Implemented modular Python software to automate hardware initialization and real-time control sequences on Raspberry Pi.

MicroMouse | IEEE (UGA) *September 2025 – March 2026*
Embedded Systems Research and Development
Team-based robotics initiative focused on designing and programming an autonomous maze-solving robot.

- Engineered a 6-channel infrared sensor polling system using dual STM32 ADCs to facilitate obstacle navigation in real time and allow wall-following for an autonomous maze-solving robot.
- Implemented a PID control algorithm based on differential error between front-inward sensors, enabling precise steering corrections and stable positioning throughout the maze.
- Managed code repository using GitHub, collaborating with the team and performing code review with VS Code's integrated Git environment.